Saturable Vacuum Electrodynamics: a tree-level vacuum-birefringence prediction for a pump–probe X-ray polarimeter

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We register, before any pump-on data exists, a falsifiable prediction for the combined optical-pump / X-ray-probe polarimeter geometry of the BIREF@HIBEF proposal. The prediction follows from a single constitutive postulate: the vacuum permittivity saturates as $\varepsilon_{\rm eff} = \varepsilon_0\, S(E)$ with the elliptic kernel $S = \sqrt{1-(E/E_{\rm c})^2}$ and one field scale $E_{\rm c} \simeq 1.13 \times 10^{17}\,{\rm V\,m^{-1}}$. Under a linearly polarized pump this yields a *tree-level*, E^2-leading birefringence $\delta n_{\rm bir} = n_\parallel - n_\perp \simeq -\frac{1}{2}(E/E_{\rm c})^2$ whose differential coefficient exceeds the one-loop QED (Euler–Heisenberg) prediction by the field-independent factor $\delta n_{\rm sat}/\delta n_{\rm QED} = 3.75\pi/\alpha^2 \simeq 2.2 \times 10^5$ (both coefficients on a single instantaneous footing). At the demonstrated ReLaX pump intensity ($10^{21}\,{\rm W\,cm^{-2}}$) the model predicts a polarization-flip probability $P_{\rm flip} \simeq 4 \times 10^{-3} - -9 \times 10^{-3}$, roughly seven orders of magnitude above the record X-ray-polarimeter purity floor (2.4×10^{-10}). The model's magnetic response keys on circulation rather than static flux, so it predicts *zero* static-magnetic-field birefringence; existing PVLAS/BMV magnetic nulls are therefore consistent by construction, and the E-route configuration is the decisive one. We state the kill criterion explicitly: a pump-on polarimetric null (at 5σ) at or below the QED co-prediction, above the demonstrated purity floor and at $\gtrsim 10^{18}\,{\rm W\,cm^{-2}}$, falsifies the model's electric sector. Because the signal sits $\sim 10^7$ above the demonstrated floor at already-demonstrated parameters, even commissioning-grade data adjudicates it. We report the coefficient honestly: the kernel *form* and the resulting coefficient *structure* are the model's content, whereas the absolute scale $E_{\rm c}$ is calibrated from the same measured constants (α, m_e) that fix the QED prediction.

I. INTRODUCTION

Quantum electrodynamics predicts that the vacuum, polarized by an intense electromagnetic field, becomes weakly birefringent [1, 2]. The effect is an α^2-suppressed one-loop response: for a probe crossing a field of strength E the two eigenmodes acquire an index difference $\delta n_{\rm QED} \sim \alpha^2(E/E_{\rm crit})^2$, with $E_{\rm crit} = m_e^2 c^3/(e\hbar) \simeq 1.3 \times 10^{18}\,{\rm V\,m^{-1}}$ the Schwinger field. The predicted signal is minute and has never been measured directly. The BIREF@HIBEF proposal [3] targets it with a pump–probe geometry at the European XFEL: a petawatt-class optical laser (ReLaX) supplies the strong field and a polarized X-ray pulse probes the induced birefringence through a high-purity crossed polarimeter.

This Letter registers a competing, tree-level prediction for the *same* observable in the *same* apparatus. It rests on one constitutive hypothesis for the vacuum's dielectric response — that the permittivity *saturates* with field strength through a specific elliptic kernel — which we treat here as a self-contained model of nonlinear electrodynamics, neutral to any wider interpretation. We call it saturable vacuum electrodynamics (SVE). The model makes a numerically definite, pre-committed prediction that differs from QED by a fixed, field-independent factor, and it carries a distinctive signature — electric-field-active but static-magnetic-field-transparent — that separates it from both QED and the Born–Infeld class of nonlinear electrodynamics [4], in which electric and magnetic responses are symmetric.

The experimental landscape frames the test. Vacuum birefringence has been sought for decades on the magnetic route — most extensively by the PVLAS experiment [5], whose null bounds nonlinear-electrodynamics parameters [6, 7] but, as we show, cannot constrain the present electric-route model. The QED signal for the XFEL pump–probe geometry has been developed in detail [8, 9], and the required high-purity X-ray polarimetry has been demonstrated with the pump off [9, 10]. The present model is distinct from these alternatives on two counts: it predicts a *tree-level* coefficient (versus QED's loop), and it is static-magnetic-transparent (versus both QED and Born–Infeld). It is also distinct from a pseudoscalar (axion-like) signal [7]: the model produces pure retardance with no parity-odd polarization rotation on the electric route, whereas an axion coupling would rotate the plane of polarization; the two are separable by the analyzer geometry.

Our aim is a clean falsifier, not a defense. We therefore (i) state the model as a two-parameter constitutive law with its one calibrated scale ledgered plainly; (ii) compute the prediction and the QED co-prediction through an identical readout chain; (iii) show that no published experiment has yet combined a strong optical pump, a polarized X-ray probe, and a polarization-flip analysis, so this is a forward prediction rather than a retrodiction; and (iv) commit, before data, to the measurement outcome that would kill the model.

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II. THE CONSTITUTIVE MODEL

A. The saturable permittivity

We postulate that the vacuum responds to an applied electric field as a dielectric whose permittivity *saturates* rather than remaining constant. Concretely, with ε_0 the low-field vacuum permittivity, we take the effective permittivity along the field to be

$$\varepsilon_{\rm eff}(E) = \varepsilon_0\, S(E), \qquad S(E) = \sqrt{1 - \left(\frac{E}{E_{\rm c}}\right)^2}, \tag{1}$$

a single elliptic (circular-arc) kernel controlled by one field scale $E_{\rm c}$. The functional form is the defining hypothesis: S is unity at low field, softens as E grows, and reaches zero at $E = E_{\rm c}$, where the medium ceases to support propagation. Equation (1) is the only nonlinearity in the model; everything below is its consequence. We take S to be a function of the lab-focus-frame field magnitude $|E|$; the single-invariant $|E|^2$ collapse (in contrast to QED's two Lorentz invariants) is a modeling choice, and its frame dependence is addressed in Sec. II below.

For a linearly polarized pump field $\mathbf{E}_0$ the response becomes uniaxial, with optic axis along $\hat{\mathbf{E}}_0$. Writing $A \equiv E/E_{\rm c}$, the probe-response tensor is the exact differential of the scalar kernel, $\varepsilon_{ij} = \varepsilon\,\delta_{ij} + 2\varepsilon' E_{0i}E_{0j}$ with $\varepsilon' = \mathrm{d}\varepsilon/\mathrm{d}(E^2)$. The two refractive eigen-indices for a probe are

$$n_\perp = (1 - A^2)^{1/4}, \qquad n_\parallel = \sqrt{\frac{1 - 2A^2}{\sqrt{1 - A^2}}}, \tag{2}$$

and the birefringence a polarimeter measures is their difference,

$$\boxed{\delta n_{\rm bir} = n_\parallel - n_\perp \;\simeq\; -\tfrac{1}{2}A^2 = -\tfrac{1}{2}\left(\frac{E}{E_{\rm c}}\right)^2} \tag{3}$$

to leading order in A. This is negative, E^2-leading, and $O(1)$ against the saturation scale $E_{\rm c}$ — it is not loop-suppressed. It vanishes at low field ($A \to 0$), recovering an unbirefringent tree-level vacuum exactly as QED does at tree level; the distinction from QED is one of *coefficient*, at the same leading power (Sec. III). Every coefficient in this paper follows from Eq. (1) by differentiation — the tensor structure, both eigen-indices, the differential and common-mode shifts, and the ratio of Sec. III; no additional parameter or coupling enters anywhere (Appendix A).

B. The one calibrated scale

The kernel (1) has one dimensionful parameter, the saturation field $E_{\rm c}$. We fix it as

$$E_{\rm c} = \sqrt{\alpha}\, E_{\rm crit}, \qquad E_{\rm crit} = \frac{m_e^2 c^3}{e\hbar}, \tag{4}$$

which evaluates to $E_{\rm c} \simeq 1.13 \times 10^{17}\,\mathrm{V\,m^{-1}}$ using the fine structure constant α and the electron mass m_e. Equation (4) places the saturation field a factor $\sqrt{\alpha}$ below the QED Schwinger field $E_{\rm crit} \simeq 1.3 \times 10^{18}\,\mathrm{V\,m^{-1}}$, and it is the origin of the field-independent ratio to QED derived below, through the exact identity

$$\left(\frac{E_{\rm crit}}{E_{\rm c}}\right)^2 = \frac{1}{\alpha}. \tag{5}$$

Honesty ledger. We separate what the model supplies from what it imports, and we hold it to the same standard as QED. *(i) The kernel form is the model's content.* That the vacuum saturates at all — possessing a tree-level, $O(1)$, birefringence-bearing structure — is the substantive, potentially wrong hypothesis. The QED vacuum has no such tree-level structure; its birefringence is an α^2-loop effect. The existence and functional form of Eq. (1) is therefore the distinguishing physics. *(ii) The coefficient structure is the model's content.* The factor $\frac{1}{2}$ in Eq. (3) and $\sqrt{\alpha}$ in Eq. (4), and hence the α^{-2} scaling of the ratio to QED, follow from the form once $E_{\rm c}$ is fixed. *(iii) The absolute scale is calibrated, not derived.* The number $E_{\rm c} \simeq 1.13 \times 10^{17}\,\mathrm{V\,m^{-1}}$ is set through Eq. (4) from the *same* measured constants (α, m_e) that every treatment of this problem imports; the model does not predict α. The resulting magnitude of the ratio to QED, $3.75\pi/\alpha^2$, is thus rooted in α in exactly the way QED's own one-loop coefficient $\sim \alpha$ is: neither framework derives α. We therefore do not present the numerical magnitude as the discriminating claim. The discriminating claim is the *existence of an $O(1)$ tree-level differential coefficient* where QED has only a loop-suppressed one — and this is what the experiment tests. *(iv) The single-footing convention, stated explicitly.* The QED Euler–Heisenberg coefficient carries a temporal-averaging convention that must be matched to the model's. The literature headline $\alpha/(15\pi)$ [Eq. (7)] is the pump's optical carrier *cycle-averaged* value ($\langle\cos^2\rangle = \frac{1}{2}$); its instantaneous (peak-field) value is twice that, $2\alpha/(15\pi)$. The model kernel $\delta n_{\rm bir} = -\frac{1}{2}(E/E_{\rm c})^2$ is, by contrast, an instantaneous algebraic response evaluated at the peak field (Appendix A). We therefore state the discriminating ratio on a *single instantaneous footing* — instantaneous model kernel against *instantaneous* one-loop coefficient $2\alpha/(15\pi)$ — giving $\delta n_{\rm sat}/\delta n_{\rm QED} = 3.75\pi/\alpha^2$ (Sec. III). Pairing instead the instantaneous kernel against the cycle-averaged $\alpha/(15\pi)$ is a mixed footing and doubles the quoted ratio to $7.5\pi/\alpha^2$; the two differ by exactly the $\langle\cos^2\rangle = \frac{1}{2}$ carrier average, and neither the order of magnitude nor any falsifier verdict of Sec. V depends on the choice. (This same $\frac{1}{2}$ carrier average is one of the two factors in the static-to-propagating QED normalization step of Sec. III.) *(v) The sector boundary was acquired, not assumed.* We record where the model failed and what that costs. *What was implicitly claimed:* the kernel of Eq. (1), written as a constitutive law with no regime boundary, reads as a universal permittivity for any electric field. *What the data ruled:* that reading is dead.

Run as a continuum static law down to atomic scales, the same kernel overshoots the muonic-hydrogen Lamb interval [11] non-perturbatively — the induced level shift exceeds the entire measured splitting, not merely its error bar (and inside a high-Z ion the kernel has no real solution at all) — with no physically-motivated short-distance cutoff to rescue it. The static-field sector of the constitutive law is therefore excluded (Sec. II, sector scope). *What survives:* the falsifier of this Letter, unchanged. The registered prediction — $P_{\rm flip}$ and the kill criterion of Sec. V — is a *radiative*-sector quantity (a propagating, weak-field, time-varying pump crossed by a probe), disjoint from the atomic static-DC sector that was excluded; the static-$\mathbf{B}$ transparency of Eq. (6) is disjoint again. Neither is touched by the exclusion. *What is open:* the *mechanism* that scopes the law to the radiative sector. That the law is radiative-sector is here a postulate; the dynamical scoping routes proposed for it — keys that would distinguish a propagating from a held field by its dynamical content — have themselves each been tested and excluded, and the static-sector response is instead derived to be charge-keyed: a held field is a real operating-point bias, so it loads. What remains open is confined to the lattice-scale regime boundary — whether the continuum law is the right accounting below one lattice pitch — and the static exclusion is robust to it: a lattice-pitch cutoff alone does not rescue the static sector. We flag this openly rather than present the constitutive law as though it carried a settled sector boundary from the outset.

The elliptic form is not a free choice. The kernel $S = \sqrt{1-(E/E_{\rm c})^2}$ is the unique response of a lossless oscillator whose amplitude is confined to a hard bound under energy conservation. If the two reactive stores exchange a conserved energy while the drive amplitude E and the residual stiffness S are complementary components of a fixed-length state ($E^2 + E_{\rm c}^2 S^2 = E_{\rm c}^2$, the Pythagorean complement — as for a saturating inductor at its flux ceiling, or the frequency pull of a pendulum swinging to a hard stop), then $S = \sqrt{1-(E/E_{\rm c})^2}$ follows and no other exponent does: a general power law $S = (1-(E/E_{\rm c})^p)^{1/p}$ satisfies the same endpoints and small-field limit for *every* p, and only the quadratic (energy) norm $p = 2$ is selected by energy conservation. The exponent is therefore a theorem of the confinement, not a fitted parameter. And it is testable in its own right: different kernel exponents predict different ratios of the anisotropic (par−perp) to the common-mode index shift, so measuring both channels fixes p — a second-order discriminator beyond the leading coefficient of Sec. III.

Family context. A saturating-field electrodynamics is not a new idea: Born and Infeld introduced one ninety years ago [4], engineering a square-root Lagrangian $\mathcal{L} \propto \sqrt{1+F/b^2}$ with a limiting field b so that the field of a point charge — and the electron self-energy — stays finite. The present kernel $S = \sqrt{1-(E/E_{\rm c})^2}$ belongs to the same saturating-field family but is a distinct branch of it: the sign under the root is reversed, so the response *softens* toward a hard ceiling $E_{\rm c}$ rather than stiffening, and (Sec. IV) the magnetic sector is not symmetric with the electric one. We situate the model here only to place its lineage; any comparison of self-energy properties across branches is outside the scope of this Letter, whose registered claim is the pump–probe birefringence coefficient.

Sector scope, stated explicitly. We assert the constitutive law of Eq. (1) for the *radiative* (pump–probe) sector only: a deep-cold, weak-field ($A^2 \simeq 6 \times 10^{-7}$), time-varying dielectric response read at optical/X-ray frequencies. We do *not* assert it as a universal static-field constitutive law extending to strong, near-nucleus atomic fields. This boundary was not assumed but tested. Before any external submission, we pre-registered a fork with a committed-in-advance outcome and computed the elliptic kernel's contribution, as a continuum static constitutive law, to the muonic-hydrogen Lamb shift. An atomic Coulomb field is not weak against $E_{\rm c}$ (at the muonic-hydrogen Bohr scale $A^2 \simeq 2.5 \times 10^{-2}$, and inside a high-$Z$ ion the field exceeds $E_{\rm c}$, where the elliptic kernel has no real solution at all), and the muonic-hydrogen $2P_{1/2}$–$2S_{1/2}$ interval is measured to 202.3706(23) meV [11]. The continuum static extension *fails this test non-perturbatively*: the induced level shift exceeds not merely the sub-μeV measurement window but the *entire* measured Lamb interval by orders of magnitude, and no lattice-scale short-distance cutoff placed at a physically motivated scale rescues it. The continuum static-field extension of Eq. (1) is therefore *excluded*; the registered radiative-sector prediction of this Letter is a separate sector, untouched by that exclusion (honesty ledger, Sec. II). The radiative scoping is at present a *postulate*: it states *that* the law is radiative-sector, not yet *why*. The dynamical scoping routes proposed for it — keys that would separate a propagating from a held field by its dynamical content — have each been tested and excluded; the static-field response is instead derived to be charge-keyed, so that a held field is a real operating-point bias and does load. The open scoping question is thereby confined to the lattice-scale regime boundary — whether the continuum form of the law is the correct accounting below one lattice pitch — and the static-sector exclusion is robust to it: a lattice-pitch cutoff alone does not rescue the static sector. The magnetic sector's circulation keying [Eq. (6), static-$\mathbf{B}$ transparency] is a distinct sector again and is unaffected by any of this.

C. The magnetic sector: static-flux transparency

The constitutive postulate is completed by a statement about the magnetic response, which is *not* symmetric with the electric one. In this model the magnetic response of the vacuum keys on the *circulation* of the field — the time-varying, curl-carrying part — rather than on static flux. A purely static magnetic field ($\partial\mathbf{B}/\partial t = 0$) does not load the magnetic response: the corresponding saturation factor stays at unity and the magnetic-route birefringence is

$$\delta n_\mu = 0 \qquad (\text{static } \mathbf{B},\ \text{exactly}). \tag{6}$$

This is a hard, parameter-free side-prediction: the model forbids static-magnetic-field vacuum birefringence at any field strength. It is the feature that most sharply separates the model from QED and from Born–Infeld-type electrodynamics, both of which predict a nonzero static-**B** birefringence. A propagating field — such as the ReLaX optical pump, for which **E** and **B** co-move with $B = E/c$ and $\partial\mathbf{B}/\partial t \neq 0$ — *does* load the response; the electric-route result of Eq. (3) already captures the full birefringence of such a wave. The consequences of Eq. (6) for existing null experiments are taken up in Sec. IV. Physically, the asymmetry follows from the response keying on the field's circulation (∂_t, curl) rather than on static flux: a static **B** carries no circulation to load, whereas the electric route and any propagating wave do.

D. Frame, dispersion, and the common-mode channel

Three points of self-consistency. *Frame.* Because S depends on the single quantity $|E|^2$, which is not a Lorentz invariant, the prediction is stated in the lab frame of the optical focus, where the pump field magnitude is defined. The relevant question is whether motion of the apparatus through any preferred frame reintroduces a spurious signal. For a laboratory static field of 2.5 T (a PVLAS-scale magnet) and a boost of $370\,\mathrm{km\,s^{-1}}$ (the Solar System's motion relative to the CMB), the motional electric field is $E \sim vB \simeq 9\times10^5\,\mathrm{V\,m^{-1}}$, giving $A^2 \sim 7\times10^{-23}$ and $\delta n \sim 3\times10^{-23}$. In the language of the Standard-Model Extension, the model belongs to the class of preferred-frame photon-sector modifications whose coefficients are experimentally bounded [12]; the estimate above shows that all current photon-sector Lorentz-invariance bounds are satisfied with orders of magnitude to spare, so the frame dependence is operationally negligible and the static-**B** transparency of Eq. (6) is not resurrected by the boost.

A sidereal signature. If the model's response frame coincides with the CMB rest frame, the preferred-frame dependence is not merely a null constraint but a positive prediction. As the Earth rotates and orbits, the projection of the $370\,\mathrm{km\,s^{-1}}$ boost onto the pump–probe geometry varies, modulating the birefringence coefficient at the fractional level $(v/c)^2 \simeq 1.5\times10^{-6}$ with sidereal and annual periods. Operationally, a pump-on flip probability measured across many sidereal days would carry $\sim 1.5\times10^{-6}$ fractional sidebands at the sidereal frequency (and its harmonics), phased to the known CMB dipole direction. This is a pre-registered *third* falsifiable signature of the model — alongside the tree-level coefficient (Sec. III) and the static-**B** transparency (Sec. IV) — accessible as a second-generation measurement once the leading signal is established. *Dispersion.* We treat the coefficient as frequency-independent (a DC-Kerr response), so the same E_c governs both the 1.55 eV pump and the ~ 10 keV probe. This is justified by scale separation: the medium's natural response scale is the electron rest energy $m_e c^2 \simeq 511$ keV, so the probe sits nearly two orders of magnitude below it, $\omega_\mathrm{probe}/\omega_0 \simeq 0.02$. A dispersive correction enters at second order in this ratio, bounding the fractional shift of the coefficient at $(\omega_\mathrm{probe}/\omega_0)^2 \sim 4\times10^{-4}$ ($< 0.1\%$) across the probe band — negligible against the $\sim 10^7$ discrimination margin, so the quasi-static coefficient is safe. *Common-mode channel.* Beyond the birefringence, the model predicts an isotropic index shift $\delta n_\mathrm{iso} = \sqrt{S} - 1 \simeq -\frac{1}{4}A^2$, common to both probe eigenmodes and hence rejected by a polarimeter. It is in principle accessible to pump–probe interferometry: the DeLLight experiment [13] is a Sagnac-interferometer measurement of exactly this common-mode vacuum index change, projecting sensitivity to the QED-level induced index (a refraction angle ~ 0.13 prad, reached at 5σ in ~ 1 month with a signal-amplification extinction factor $\mathcal{F} = 0.4\times10^{-5}$). The present model's isotropic shift at $10^{21}\,\mathrm{W\,cm^{-2}}$ is $\sim 1.5\times10^{-7}$ over the focus. We do not register a prediction for the common-mode channel here; the falsifier of this Letter is the polarimetric differential, not the interferometric common mode.

III. THE PREDICTION

A. The matched differential and the readout chain

A birefringence polarimeter measures the *difference* $n_\parallel - n_\perp$ between the two eigenmodes; the common-mode index shift shared by both modes is rejected by the instrument. The falsifier observable is therefore the differential δn_bir of Eq. (3), and any comparison with QED must be made differential-against-differential. For a probe crossing a strong propagating field the one-loop Euler–Heisenberg birefringence is [1, 8, 9]

$$\delta n_\mathrm{QED} = \frac{\alpha}{15\pi}\left(\frac{E}{E_\mathrm{crit}}\right)^2, \tag{7}$$

the coefficient $\alpha/(15\pi)$ being the head-on ($\vartheta_\mathrm{coll} = \pi$), optimal-polarization value for the counter-propagating pump–probe geometry of the BIREF@HIBEF conventional scenario [3, 8], in which the polarization-flip signal is maximized (it scales as $(1-\cos\vartheta_\mathrm{coll})^4$). We anchor it to two external arbiters. First, the magnetic-route differential $\delta n = 3A_e B^2$ with the measured vacuum magnetic-birefringence constant $A_e = 1.32\times10^{-24}\,\mathrm{T^{-2}}$ [5, 6]: at the $E \leftrightarrow cB$ duality point the leading $(E^2 - B^2)^2$ invariant fixes the static-field (DC) coefficient at $3A_e B_\mathrm{crit}^2 = \alpha/(30\pi)$. The step from this static endpoint to the headline Eq. (7) is a net factor of two that factorizes into a factor of four from the head-on collision geometry (the birefringence of a probe crossing a strong pump lives in the pump–probe cross terms, not in the single on-shell plane wave, for which both Euler–Heisenberg invariants vanish), $\alpha/(30\pi) \to 2\alpha/(15\pi)$, times a $\langle\cos^2\rangle = \frac{1}{2}$ carrier average, $2\alpha/(15\pi) \to \alpha/(15\pi)$ — the static endpoint is DC, the headline is cycle-averaged, so it is not a pure geometry factor. Second, the same coefficient reproduces the focus-integrated signal estimate of the BIREF@HIBEF proposal [3] at its stated parameters. Both Eq. (3) and Eq. (7) are E^2-leading: an E^2 slope alone does *not* discriminate the two. The discriminator is the coefficient.

For a single-pass geometry (no cavity build-up), an index difference $\delta n_{\rm bir}$ over an interaction length z imprints a retardance $\Delta\varphi = (2\pi/\lambda)|\delta n_{\rm bir}|\,z$ on a probe of wavelength λ, rotating a fraction

$$P_{\rm flip} = \sin^2\left(\frac{\Delta\varphi}{2}\right) \tag{8}$$

of the probe intensity into the crossed polarimeter channel. This bounded form reduces to the perturbative $(\Delta\varphi/2)^2$ when $\Delta\varphi \ll 1$. We compute both the model and the QED co-prediction through this *identical* chain, changing only $\delta n_{\rm bir}$, so the comparison is like-for-like and free of a strawman baseline. *Field convention.* The field E entering Eqs. (3) and (7) throughout, and in Table I, is the peak carrier amplitude $E = \sqrt{2I/(c\varepsilon_0)}$ of the optical pump at intensity I; both legs are evaluated at this same peak field through the identical chain. Two distinct statements about this choice must be kept apart. Merely *re-parametrizing* the field — e.g. substituting the cycle-averaged $\langle E^2\rangle = E^2/2$ into the same two formulas — rescales both legs' absolute retardance by a common factor and leaves the ratio of Sec. III untouched. That is not the same as placing both legs' *coefficients* on a single temporal footing (instantaneous vs. cycle-averaged), which does move the numerical ratio by an $O(1)$ factor; this is the convention item ledgered in Sec. II item (iv), and it changes no order of magnitude and no falsifier verdict.

B. The field-independent ratio

Because both responses are E^2-leading, their ratio is independent of field strength. The ratio is stated on a *single temporal footing*: both coefficients are the instantaneous (peak-field) response, so the model kernel $\delta n_{\rm bir} = -\frac{1}{2}(E/E_{\rm c})^2$ is paired with the *instantaneous* one-loop coefficient $2\alpha/(15\pi)$ [the cycle-averaged headline of Eq. (7) is half of it; Sec. II item (iv)]. Using Eqs. (3), the instantaneous form of (7), and the identity (5),

$$\frac{\delta n_{\rm sat}}{\delta n_{\rm QED}} = \frac{\frac{1}{2}}{2\alpha/15\pi}\left(\frac{E_{\rm crit}}{E_{\rm c}}\right)^2 = \frac{15\pi}{4\alpha^2} = \frac{3.75\pi}{\alpha^2} \simeq 2.2\times 10^5. \tag{9}$$

The model predicts a differential coefficient roughly five orders of magnitude larger than QED's, at every field. As stressed in the honesty ledger, the *value* $3.75\pi/\alpha^2$ is α-rooted — as is QED's own coefficient — and is not itself the claim; the claim is the existence of an $O(1)$ tree-level differential where QED has only a loop. (A prior draft quoted $7.5\pi/\alpha^2 \simeq 4.4\times 10^5$, which paired the instantaneous model kernel against the *cycle-averaged* one-loop coefficient — a mixed footing; the single-footing value above is a factor of two smaller, and no order of magnitude or falsifier verdict changes. See Sec. II item (iv).)

C. Per-scenario predictions

Table I freezes the prediction for the scenarios of the BIREF@HIBEF proposal [3]: the probe photon energies of the conventional, dark-field, and high-energy configurations, at the demonstrated ReLaX pump intensity and at the higher design intensities, through the single-pass chain of Eqs. (8) with $z = 10\,\mu$m. At the *demonstrated* pump (10^{21} W cm^{-2}, $E \simeq 8.7\times 10^{13}$ V m^{-1}, $A^2 \simeq 5.9\times 10^{-7}$) the retardance is small ($\Delta\varphi/2 \lesssim 0.1$) and the flip probability is a well-behaved $P_{\rm flip} \simeq 4\times 10^{-3}$ – -9×10^{-3} across the three probe energies — no non-perturbative regime is invoked. The QED co-prediction through the same single-pass chain (instantaneous footing, Sec. III) is $\sim 1\times 10^{-13}$, so the two hypotheses are separated by ~ 11 orders of magnitude in flip probability. We note that this single-pass, flat-z chain gives a deliberately conservative QED baseline: the proposal's [3] focus-integrated estimate is $\sim 10^{-12}$ at the same pump, several orders of magnitude higher. Focus integration raises the model and QED legs by the *same* geometric factor, so the field- and geometry-independent ratio of Eq. (9) — the actual discriminator — is unchanged, and the model margin above the polarimeter floor only improves.

IV. THE SIGNATURE AND PRIOR NULLS

The model's identifying signature is that it is *electric-route active but static-magnetic-route transparent* [Eq. (6)]. This distinguishes it from both of the standard alternatives. QED predicts a nonzero static-$\mathbf{B}$ vacuum birefringence, and the entire PVLAS/BMV program was built to detect exactly that; Born–Infeld-type nonlinear electrodynamics [4] likewise predicts symmetric electric and magnetic responses. The present model predicts none in a static magnetic field.

A consequence is that the existing magnetic-route nulls are consistent with this model *by construction*, not by smallness of a coefficient. The PVLAS experiment, in a 25-year effort, measured no vacuum magnetic birefringence at a sensitivity approaching the QED prediction [5]. In this model that null is required: $\delta n_\mu = 0$ exactly for static $\mathbf{B}$. The magnetic-route facilities therefore neither support nor test the present prediction; they confirm the side-prediction of Eq. (6) and leave the electric route open.

The electric route requires a strong *propagating* field crossed by a polarized probe — precisely the XFEL-plus-

TABLE I. Frozen per-scenario predictions for the BIREF@HIBEF probe energies, computed through the single-pass readout chain [Eqs. (3), (7), (8)] at $z = 10\,\mu$m. Both legs are evaluated on the single instantaneous (peak-field) footing of Sec. III, so the QED column uses the instantaneous one-loop coefficient $2\alpha/(15\pi)$ and the model/QED column equals $(3.75\pi/\alpha^2)^2$ in the small-angle rows; the model flip probability is unchanged from the peak-field kernel. The three demonstrated-pump rows are the bankable prediction. The two design rows are shown for context: the $10^{22}\,\mathrm{W\,cm^{-2}}$ row is saturating (retardance $\Delta\varphi = 1.47$ rad, so the small-angle approximation no longer holds and the exact $\sin^2$ value is quoted), and the $10^{23}\,\mathrm{W\,cm^{-2}}$ row (retardance $\Delta\varphi = 14.7$ rad, many-wrap) is a named validity limit of the single-pass mapping and is *not* part of the registered prediction (Sec. V).

Scenario (probe)	Pump [$\mathrm{W\,cm^{-2}}$]	E [$\mathrm{V\,m^{-1}}$]	A^2	P_{flip} (model)	P_{flip} (QED)	model/QED	regime
Conventional (9835 eV)	10^{21} (demonstrated)	8.68×10^{13}	5.90×10^{-7}	5.39×10^{-3}	1.10×10^{-13}	4.89×10^{10}	small-angle
Dark-field (8766 eV)	10^{21} (demonstrated)	8.68×10^{13}	5.90×10^{-7}	4.28×10^{-3}	8.76×10^{-14}	4.89×10^{10}	small-angle
High-energy (12 914 eV)	10^{21} (demonstrated)	8.68×10^{13}	5.90×10^{-7}	9.28×10^{-3}	1.90×10^{-13}	4.88×10^{10}	small-angle
Conventional (9835 eV)	10^{22} (design)	2.74×10^{14}	5.90×10^{-6}	4.49×10^{-1}	1.10×10^{-11}	4.07×10^{10}	saturating
Conventional (9835 eV)	10^{23} (design)	8.68×10^{14}	5.90×10^{-5}	$7.65 \times 10^{-1\dagger}$	1.10×10^{-9}	—	*beyond validity*†

† Excluded from the registered prediction: at $10^{23}\,\mathrm{W\,cm^{-2}}$ the single-pass retardance is $\Delta\varphi \simeq 14.7$ rad (many-wrap); the naive perturbative flip probability there exceeds unity ($\simeq 54$), so the bounded $\sin^2$ value masks a broken single-pass mapping. See Sec. V.

optical-laser geometry. Theory for the QED signal in this configuration is well developed [3, 8]; the same geometry is the uniquely decisive test of the present model, because it is the one configuration in which the model and QED both predict a nonzero, matched, differential signal and differ only in coefficient.

V. FALSIFIABILITY

We commit, before any pump-on data, to the following outcome as decisive.

Kill criterion. A pump-on polarimetric run at $\gtrsim 10^{18}\,\mathrm{W\,cm^{-2}}$, with a polarimeter operating at or below the demonstrated purity floor, that measures a polarization-flip probability consistent with zero (or with the QED co-prediction) at 5σ — i.e. an upper limit $P_{\mathrm{flip}} < 10^{-8}$, still $\sim 10^6$ below the model's prediction and $\sim 10^5$ above the QED level — falsifies the electric sector of this model. This is a QED-sized differential coefficient where the model demands an $O(1)$ one, and we record it as a clean falsification: the model's electric-route hypothesis is closed, with no rescue. Conversely, a measured flip probability of the predicted size ($4 \times 10^{-3} - -9 \times 10^{-3}$ at the demonstrated pump, some seven orders of magnitude above the floor) is inconsistent with QED at this observable.

Why commissioning data suffices. The record X-ray-polarimeter purity is 8×10^{-11} [9], and a single-measurement demonstrated floor of 2.4×10^{-10} was reached earlier [10]; both were measured with the pump off. Taking the conservative 2.4×10^{-10} floor, at the demonstrated ReLaX intensity the model predicts $P_{\mathrm{flip}} \simeq 4 \times 10^{-3} - -9 \times 10^{-3}$ (Table I), a factor $\sim 2 \times 10^7$ above it. The prediction is thus not a precision measurement at the edge of feasibility: it is a $\sim 10^7$ effect at already-demonstrated parameters, so even a commissioning-grade pump-on polarimetric measurement adjudicates it.

Figure 1 places the prediction, the QED co-prediction, the record floor, and the demonstrated pump on the exposure plane.

Systematics and reanalysis. Converting a crossed-polarimeter count ratio to a flip probability requires accounting for the instrumental factors the proposal enumerates: diamond/analyzer reflection losses ($\sim 2\%$ per reflection), lens and optics transmission, analyzer acceptance bandwidth, and beam-overlap and timing jitter, together with pump-induced backgrounds (plasma emission, scattered pump light) that could mimic or mask a signal in the crossed channel [3, 9]. Each of these is an $O(1)$–$O(10)$ correction; none approaches the $\sim 10^7$ margin between the model prediction and the floor, so the kill test is robust to the full instrumental budget. Operationally, a third party can adjudicate this prediction from a published pump-on run by extracting the crossed-channel fraction $N_\perp/N$, correcting it for the enumerated throughput and background factors, and comparing the result and its error bar against the frozen entries of Table I: a value at the 10^{-3} level confirms the model, a 5σ upper limit below 10^{-8} falsifies it.

This is a forward prediction. No published experiment has combined a strong optical pump, a polarized X-ray probe, and a polarization-flip analysis at the required geometry. The reported 8×10^{-11} [9] and 2.4×10^{-10} [10] polarimeter purities were measured with the pump off; the March-2024 HIBEF beamtime was X-ray-only, with the optical pump not fired [3]; the all-optical and X-ray–X-ray light-by-light searches, the heavy-ion measurements, and the PVLAS magnetic-route null [5] all lack the combination of a polarized X-ray probe passing through a strong optical focus with flip analysis. The exposure region that would bound the model's E-route signal is therefore empty (Fig. 1), and the present table is a pre-registration rather than a retrodiction.

An honest exclusion. The highest design intensity of the proposal, $10^{23}\,\mathrm{W\,cm^{-2}}$, drives the single-pass retardance to $\Delta\varphi \simeq 14.7\,\mathrm{rad}$ (Table I). In this many-wrap regime the crossed-polarimeter flip fraction of Eq. (8) is a rapidly oscillating function whose single-pass interpretation is not credible. We do *not* register a prediction there; it is a named validity limit of the readout mapping, not part of the falsifier. The registered prediction lives entirely at the demonstrated-pump scenarios, where the small-angle mapping holds.

Consistency with existing bounds. The same calibrated scale E_c leaves the model consistent with precision optical measurements that do not use the pump–probe geometry. The implied vacuum Kerr coefficient $n_2 = 1/(2c\varepsilon_0 E_\mathrm{c}^2) \simeq 1.5 \times 10^{-32}\,\mathrm{m^2\,W^{-1}}$ gives a self-focusing critical power $\sim 6.5 \times 10^3\,\mathrm{PW}$, far above any existing laser, so no vacuum self-focusing is expected (a propagating-field, and hence radiative-sector, bound); the circulating fields of gravitational-wave interferometers produce a retardance $\sim 10^{-11}$ rad, far below their sensitivity (again a propagating-field bound); and, as shown in Sec. II, *weak* laboratory static fields are unaffected by the preferred-frame boost (the motional field they induce is $A^2 \sim 7 \times 10^{-23}$). This last is a bound on weak macroscopic static fields only; it is *not* a claim that the constitutive law extends into the strong, near-nucleus atomic static sector, which is separately excluded (Sec. II, sector scope). The interferometric route that would probe the common-mode index — the DeLLight experiment [13], projecting sensitivity to the QED-level index change — has not yet reported a pump-on measurement in this regime, so it too leaves the model unconstrained at present. The model is thus not already excluded by any published precision-optics bound [7, 13].

Pre-registration. The predictions of Table I are fixed before any BIREF@HIBEF pump-on collision data exist. Their priority is anchored by an OpenTimestamps proof: the SHA-256 hash of the frozen prediction is committed to the Bitcoin blockchain, so the hash (and hence the existence of a prediction fixed at that block time) is public while the manuscript content remains private until disclosure. No entry is revised after pump-on data become available, and any revised prediction would carry its own later timestamp.

VI. CONCLUSION

A single constitutive postulate — a vacuum permittivity that saturates through the elliptic kernel of Eq. (1), with one calibrated field scale — yields a definite, pre-registered, tree-level birefringence prediction for the pump–probe X-ray polarimeter geometry of BIREF@HIBEF. The prediction exceeds the QED co-prediction by the field-independent factor $3.75\pi/\alpha^2$ (both coefficients on a single instantaneous footing; Sec. III), sits $\sim 10^7$ above the demonstrated polarimeter floor at already-demonstrated pump parameters, and carries a hard side-prediction — no static-magnetic-field vacuum birefringence — that renders the existing PVLAS/BMV

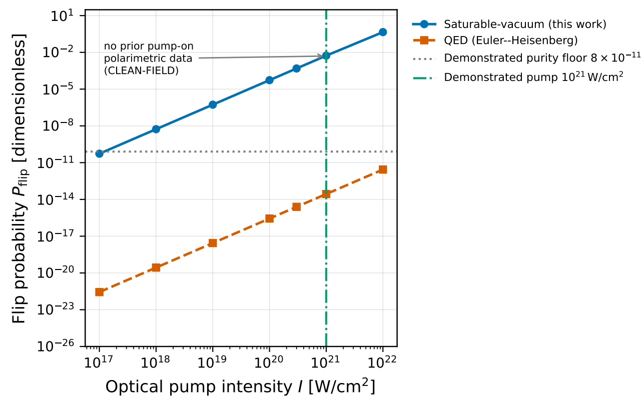

FIG. 1. The exposure plane. Predicted polarization-flip probability P_flip versus optical-pump intensity for the saturable-vacuum model (blue, Eqs. (3) and (8)) and for the QED Euler–Heisenberg co-prediction (orange, Eq. (7)), through the identical single-pass readout chain at probe energy 9835 eV, $z = 10\,\mu\mathrm{m}$. Both scale as intensity squared, so their ratio is field-independent. The dotted line marks the record X-ray-polarimeter purity (8×10^{-11}, Ref. [9]; the conservative 2.4×10^{-10} single-measurement floor of Ref. [10] used for the kill criterion lies just above it), and the dash-dot line the demonstrated ReLaX pump ($10^{21}\,\mathrm{W\,cm^{-2}}$). No prior experiment occupies the pump-on polarimetric exposure region, so the model's prediction there is unconstrained by existing data.

nulls consistent by construction and singles out the electric-route XFEL geometry as decisive. We have stated the kill criterion in advance and ledgered the one imported scale plainly. The measurement that decides the model is within reach of commissioning-grade data at an already-operating facility.

Appendix A: Derivation of the eigen-indices

The eigen-indices of Eq. (2) follow from the constitutive law Eq. (1) by direct differentiation; we display every step. The scalar law relates displacement and field through the field-magnitude invariant $u \equiv |E|^2$,

$$D_i = \varepsilon_0\, S(u)\, E_i, \qquad S(u) = \sqrt{1 - u/E_\mathrm{c}^2}. \tag{A1}$$

A weak probe $\mathbf{e}$ rides on a strong, linearly polarized pump $\mathbf{E}_0$. Its small-signal response is the permittivity tensor $\varepsilon_{ij} = \partial D_i/\partial E_j$ evaluated at the pump. Applying the product rule to Eq. (A1),

$$\varepsilon_{ij} = \frac{\partial}{\partial E_j}\left[\varepsilon_0 S(u) E_i\right] = \varepsilon_0\Big(S\,\frac{\partial E_i}{\partial E_j} + E_i\,\frac{\partial S}{\partial E_j}\Big), \qquad \frac{\partial E_i}{\partial E_j} = \delta_{ij}, \tag{A2}$$

and the chain rule fixes the second term through $\partial S/\partial E_j = S'(u)\,\partial u/\partial E_j = 2S'(u)E_j$, so that evaluated at the pump

$$\varepsilon_{ij} = \frac{\partial}{\partial E_j}\left[\varepsilon_0 S(u) E_i\right]\Big|_{\mathbf{E}_0} = \varepsilon_0 S(u_0)\,\delta_{ij} + 2\varepsilon_0 S'(u_0)\,E_{0i}E_{0j}, \tag{A3}$$

with $u_0 = |\mathbf{E}_0|^2 \equiv E^2$ and $S' = \mathrm{d}S/\mathrm{d}u$. This is uniaxial, with optic axis along $\hat{\mathbf{E}}_0$: the two eigenvalues are $\varepsilon_0 S$ (transverse, twofold) and $\varepsilon_0(S + 2S'E^2)$ (longitudinal). Differentiating the kernel gives, with $A \equiv E/E_\mathrm{c}$,

$$S'(u) = -\frac{1}{2E_\mathrm{c}^2\sqrt{1-u/E_\mathrm{c}^2}} = -\frac{1}{2E_\mathrm{c}^2 S}, \tag{A4}$$

so that $2S'(u_0)\,E^2 = -A^2/S$. The probe indices are the square roots of the eigenvalues,

$$n_\perp = \sqrt{S} = (1-A^2)^{1/4}, \tag{A5}$$

$$n_\parallel = \sqrt{S - \frac{A^2}{S}} = \sqrt{\frac{S^2 - A^2}{S}} = \sqrt{\frac{1-2A^2}{\sqrt{1-A^2}}}, \tag{A6}$$

which are Eq. (2). Expanding for $A \ll 1$,

$$n_\perp \simeq 1 - \tfrac{1}{4}A^2, \qquad n_\parallel \simeq 1 - \tfrac{3}{4}A^2, \tag{A7}$$

so the birefringence (the polarimeter observable) and the common-mode shift (the polarimeter rejects it) are

$$\delta n_\mathrm{bir} = n_\parallel - n_\perp \simeq -\tfrac{1}{2}A^2, \qquad \delta n_\mathrm{iso} = n_\perp - 1 \simeq -\tfrac{1}{4}A^2, \tag{A8}$$

the differential being exactly twice the common-mode shift. The ratio of Sec. III then follows from Eq. (A8) and the QED coefficient alone; no parameter beyond E_c enters at any step.

[1] W. Heisenberg and H. Euler, Z. Phys. **98**, 714 (1936).
[2] J. Schwinger, Phys. Rev. **82**, 664 (1951).
[3] N. Ahmadiniaz, C. Bähtz, A. Benediktovitch, C. Bömer, L. Bocklage, T. E. Cowan, H. Gies, S. Göde, J. Grenzer, T. Heinzl, F. Karbstein, B. King, T. Kluge, B. Marx-Glowna, M. Nakatsutsumi, G. G. Paulus, R. Röhlsberger, N. Rohringer, R. Sauerbrey, H.-P. Schlenvoigt, S. M. Schmidt, R. Schützhold, U. Schramm, D. Seipt, T. Stöhlker, T. Toncian, U. Zastrau, and M. Zepf, High Power Laser Sci. Eng. **13**, e7 (2025), arXiv:2405.18063.
[4] M. Born and L. Infeld, Proc. R. Soc. Lond. A **144**, 425 (1934).
[5] A. Ejlli, F. Della Valle, U. Gastaldi, G. Messineo, R. Pengo, G. Ruoso, and G. Zavattini, Phys. Rep. **871**, 1 (2020).
[6] R. Battesti and C. Rizzo, Rep. Prog. Phys. **76**, 016401 (2013).
[7] M. Fouché, R. Battesti, and C. Rizzo, Phys. Rev. D **93**, 093020 (2016).
[8] F. Karbstein, H. Gies, M. Reuter, and M. Zepf, Phys. Rev. D **92**, 071301 (2015).
[9] F. Karbstein, C. Sundqvist, K. S. Schulze, I. Uschmann, H. Gies, and G. G. Paulus, New J. Phys. **23**, 095001 (2021).
[10] B. Marx, K. S. Schulze, I. Uschmann, T. Kämpfer, R. Lötzsch, O. Wehrhan, W. Wagner, C. Detlefs, T. Roth, J. Härtwig, E. Förster, T. Stöhlker, and G. G. Paulus, Phys. Rev. Lett. **110**, 254801 (2013).
[11] A. Antognini, F. Nez, K. Schuhmann, F. D. Amaro, F. Biraben, J. M. R. Cardoso, D. S. Covita, A. Dax, S. Dhawan, M. Diepold, L. M. P. Fernandes, A. Giesen, A. L. Gouvea, T. Graf, T. W. Hänsch, P. Indelicato, L. Julien, C.-Y. Kao, P. Knowles, F. Kottmann, E.-O. Le Bigot, Y.-W. Liu, J. A. M. Lopes, L. Ludhova, C. M. B. Monteiro, F. Mulhauser, T. Nebel, P. Rabinowitz, J. M. F. dos Santos, L. A. Schaller, C. Schwob, D. Taqqu, J. F. C. A. Veloso, J. Vogelsang, and R. Pohl, Science **339**, 417 (2013).
[12] V. A. Kostelecký and N. Russell, Rev. Mod. Phys. **83**, 11 (2011).
[13] S. Robertson, A. Mailliet, X. Sarazin, F. Couchot, E. Baynard, J. Demailly, M. Pittman, A. Djannati-Ataï, S. Kazamias, and M. Urban, Phys. Rev. A **103**, 023524 (2021), arXiv:2011.13889.